



研究背景

在高维数据中，通常包含一些无关冗余特征，基于包裹法的特征选择问题中，连续优化算法做离散化后容易出现搜索方向不稳定、特征位反复翻转的情况。

研究背景

基于SCA：结构简单，离散特征选择时依赖单一最优解。

研究目标：在使用统一的分类器与二值化基础上，利用有限的评估预算，找到分类性能与特征稀疏性更均衡的特征子集。

问题分解：搜索个体“从哪里出发、朝哪里搜索、移动多大”。

已有研究

定义连续分数，提出连续函数选 / 不选？

反向学习、邻域搜索、变异和折射学习，记忆引导策略
时间变化的转换函数、便宜的代理模型、多目标优化。

Wrapper-base 通用性？

数据集单一，记忆引导受到Attention的影响。

停止步骤需要根据不同的数据集来调试，对数据集的依赖性强，可以属于超超参数。

代理模型判断易错；多目标最后需要人为维护多个候选方案需要更多计算。

核心：不改变分类器而是提供搜索方案。

本研究工作与贡献

标准SCA：

$$x_{i,j}^{t+1} = \begin{cases} x_{i,j}^t + r_1^t \sin(r_{2,i,j}^t) \left| r_{3,i,j}^t g_j^t - x_{i,j}^t \right|, & r_{4,i,j}^t < 0.5, \\ x_{i,j}^t + r_1^t \cos(r_{2,i,j}^t) \left| r_{3,i,j}^t g_j^t - x_{i,j}^t \right|, & r_{4,i,j}^t \geq 0.5. \end{cases}$$

第一，构建了短期分数阶记忆，避免每次只根据当前状态更新。

$$x_{\text{mem},i}^t = - \sum_{k=1}^3 \omega_k^{(\alpha)} x_i^{t+1-k} = c_0 x_i^t + c_1 x_i^{t-1} + c_2 x_i^{t-2}, \quad (13)$$

where the historical coefficients satisfy $c_j = -\omega_{j+1}^{(\alpha)}$. Therefore,

$$c_0 = \alpha, \quad c_1 = \frac{1}{2}\alpha(1-\alpha), \quad c_2 = \frac{1}{6}\alpha(1-\alpha)(2-\alpha), \quad 0 < \alpha < 1. \quad (14)$$

第二，动态精英引导。由多个优秀解加权提供方向，避免过早依赖单一最优解，后期再逐渐收缩到全局最优。

$$L^t = \max(1, \text{round}[L_0 - (L_0 - 1)\tau]), \quad L_0 = \max\left(1, \text{round}\left(\frac{N}{2}\right)\right), \quad (16)$$

$$\rho_k^t = \frac{f_{\text{worst}}^t - f(x_{q_k}^t)}{\sum_{\ell=1}^{L^t} [f_{\text{worst}}^t - f(x_{q_\ell}^t)]}, \quad k = 1, 2, \dots, L^t. \quad \mathbf{E}^t = \sum_{k=1}^{L^t} \rho_k^t x_{q_k}^t,$$

第三，构建了一个 α 耦合的快衰减幅度，让搜索更快衰减，减少后期无效扰动。（其他研究已经证明的可行性，非核心的创新，消融实验中也做了这个变体）。

$$\mathbf{G}^t = (1 - \eta^t) \mathbf{E}^t + \eta^t \mathbf{g}^t, \quad \eta^t = \tau^{3/2}, \quad (21)$$

$$\hat{x}_{i,j}^{t+1} = x_{\text{mem},i,j}^t + r_1^t \psi_{i,j}^t \left| r_{3,i,j}^t G_j^t - x_{i,j}^t \right|, \quad (26)$$

$$\psi_{i,j}^t = \begin{cases} \sin(r_{2,i,j}^t), & r_{4,i,j}^t < 0.5, \\ \cos(r_{2,i,j}^t), & r_{4,i,j}^t \geq 0.5, \end{cases} \quad (27)$$

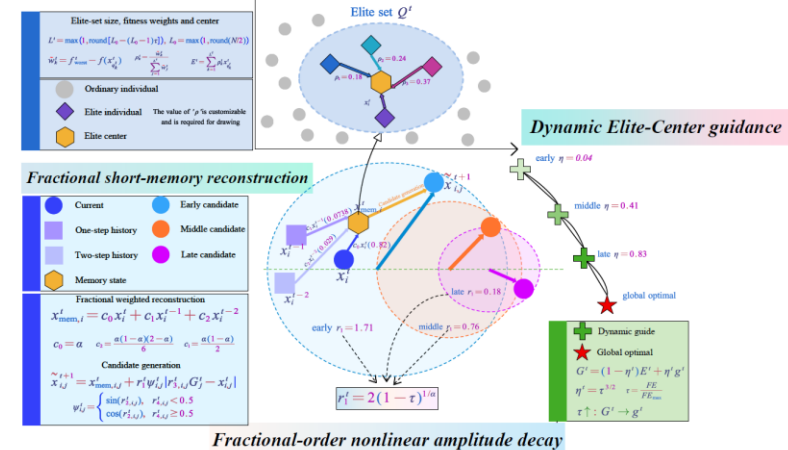
特征通过 0.5 阈值二值化，目标函数为： $f = 0.99 \text{MCE}_{CV} + 0.01 \frac{|S|}{D}$.

Author Contributions: Conceptualization, S.G.; Methodology, S.G.; Software, Y.X.; Validation, Y.X. and W.L.; Formal analysis, W.L.; Investigation, Y.X. and K.X.; Resources, Y.X. and W.L.; Data curation, W.L. and K.X.; Writing—original draft, Y.X.; Writing—review and editing, S.G. and B.Q.; Visualization, Y.X.; Supervision, S.G. and B.Q.; Project administration, S.G. and B.Q.; Funding acquisition, S.G. and B.Q. All authors have read and agreed to the published version of the manuscript.

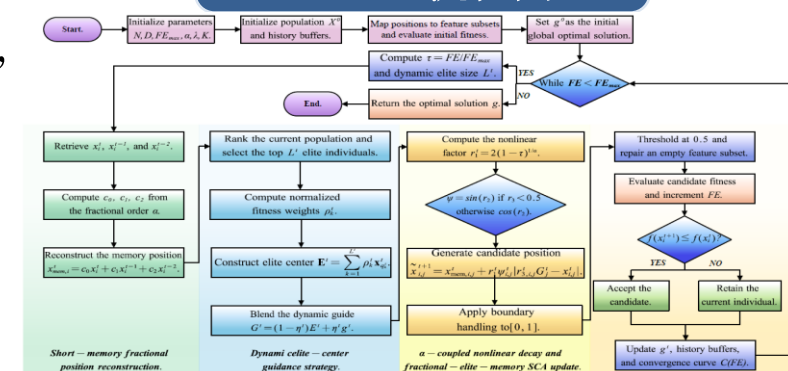
Funding: This work was supported by the National Natural Science Foundation of China under Grant No. 62376109 and the National Undergraduate Innovation and Entrepreneurship Training Program of Jiangsu University of Science and Technology (Project No. X2026102890014).

Data Availability Statement: The data used in this study must be authentic and reliable to ensure the validity and reproducibility of the experimental results. The datasets and source code used in this study are openly available at: <https://github.com/xieix-code/FOSCA> (Accessed on June 24, 2026).

FOSCA机制图



FOSCA流程图



FOSCA伪代码

Algorithm 1 Pseudo-code of FOSCA.
Require: Population size N , maximum function evaluations $FE_{\max}$, dimension D , bounds lb and ub , fitness function $f(\cdot)$, fractional order α .
Ensure: Best solution $\mathbf{g}$, best fitness $f(\mathbf{g})$, convergence curve C .
1: Initialize N individuals $\mathbf{x}_i^0 \in [lb, ub]^D$, $i = 1, 2, \dots, N$.
2: Set the historical caches $\mathbf{x}_i^{-1} = \mathbf{x}_i^0$ and $\mathbf{x}_i^{-2} = \mathbf{x}_i^0$.
3: Evaluate each individual using the fitness function in Equation (6) and record the initial global optimal solution $\mathbf{g}$.
4: **while** $FE < FE_{\max}$ **do**
5: Compute the search progress $\tau = FE / FE_{\max}$.
6: Determine the number of elite individuals L^t using Equation (16).
7: Construct the elite center $\mathbf{E}^t$ using Equation (19) and (20).
8: Generate the dynamic guide $\mathbf{G}^t$ using Equation (21).
9: Compute the fractional memory position $\mathbf{x}_{\text{mem},i}^t$ using Equation (13).
10: Compute the alpha-coupled search factor r_1^t using Equation (22).
11: Generate candidate individuals using Equation (26) and (27).
12: Apply boundary handling using Equation (28).
13: **for** $i = 1$ to N **do**
14: Evaluate candidate individuals using Equation (6) and update FE .
15: Apply greedy acceptance using Equation (29).
16: Update the global optimal solution using Equation (30).
17: Record the best-so-far fitness in $C(FE)$.
18: **end for**
19: Update the historical position caches.
20: **end while**
21: **return** Return the optimal solution $\mathbf{g}$, its fitness value $f(\mathbf{g})$ and its convergence curve C .

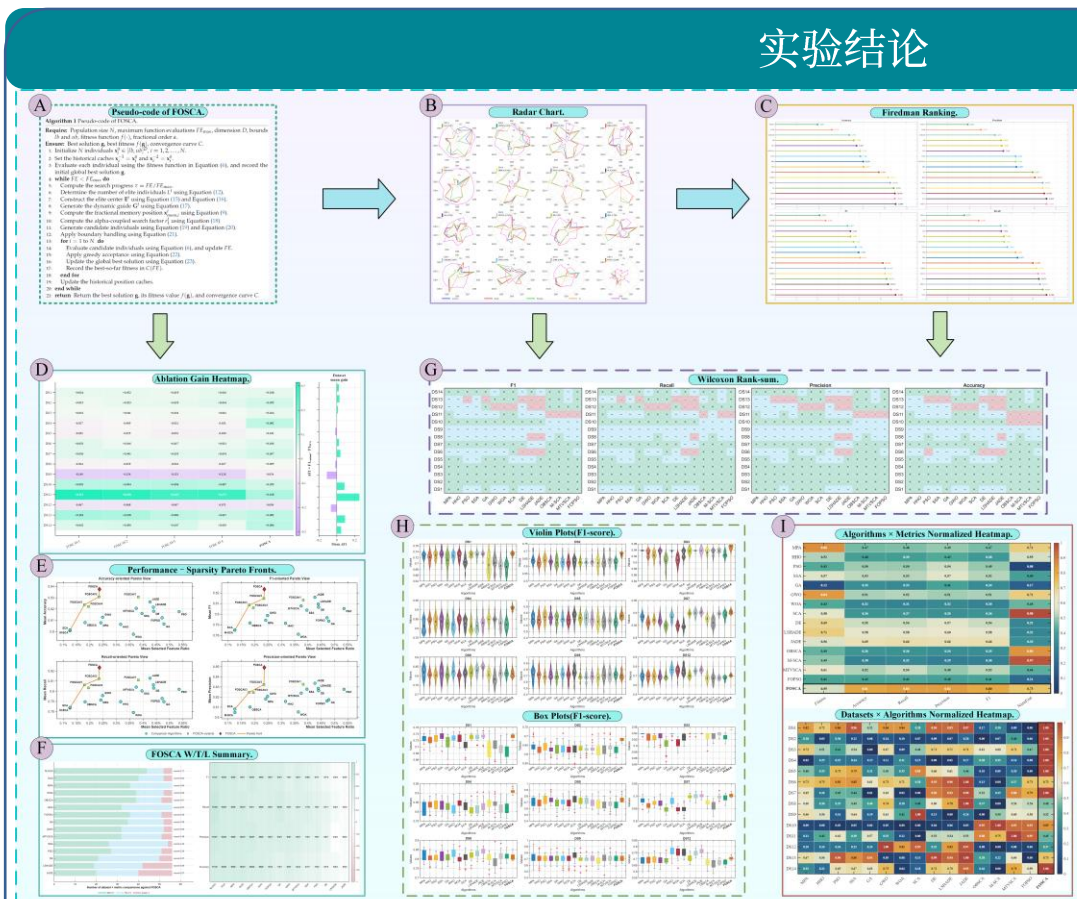
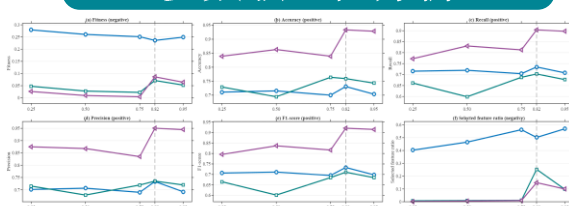
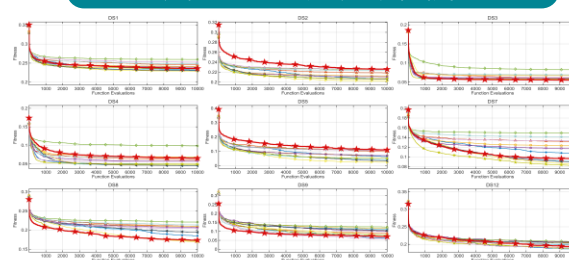


Fig. 1 The graphical abstract is organized into nine complementary components: (A) Pseudo-code of FOSCA; (B) FOSCA and other algorithms’ Firedman ranking radar chart on each dataset; (C) Firedman ranking of FOSCA compared to other algorithms; (D) FOSCA Variants Ablation Gain Heatmap; (E) Performance–Sparsity Pareto Fronts across 14 Datasets; (F) FOSCA Wilcoxon rank-sum test results chart: W/T/L summation and comparison algorithm; (G) FOSCA pairwise Wilcoxon outcome symbols across datasets; (H) FOSCA F1-score stability analysis (Violin plots and Box plots); (I) Algorithms / Metrics / Datasets pairwise Normalized Heatmap;

参数敏感性分析



算法收敛性分析



实际CPU运行时间（8并）

